EAS 6134: Inverse Methods in EAS

Underdetermined Least Squares (UDLS) & Data Fitting

2 February 2026

Overdetermined Least Squares (ODLS)

Three ways to solve

1. We solved ODLS problem by direct unconstrained minimization to get the **normal equations** — simple but not necessarily stable.

2. Next we saw how to solve ODLS by **QR decomposition of A**, and back substitution — very stable. **A=QR** works for tall matrices $m \geq n$

3. Solve as a **Lagrange multiplier problem** minimizing $||\mathbf{r}||^2$ and treating data as constraints — good for sparse design matrix **A**.

UDLS with uncertainties -we don't want to fit noisy data exactly

In our first approach to regularization we were solving the exact problem $\mathbf{Ax} = \mathbf{y}$ with minimum $||\mathbf{x}||^{\mathbf{2}}$.
What about fitting the observations to some reasonable misfit level γ?

$$||\mathbf{Ax} \ - \mathbf{y}|| \leq \gamma \tag{28}$$

For the time being we'll concentrate on the equality case in (28), and now let's suppose we want to minimize

$$f(\mathbf{x}) = ||\mathbf{Px}||^{\mathbf{2}} \tag{29}$$

UDLS with uncertainties
we don't want to fit noisy data exactly

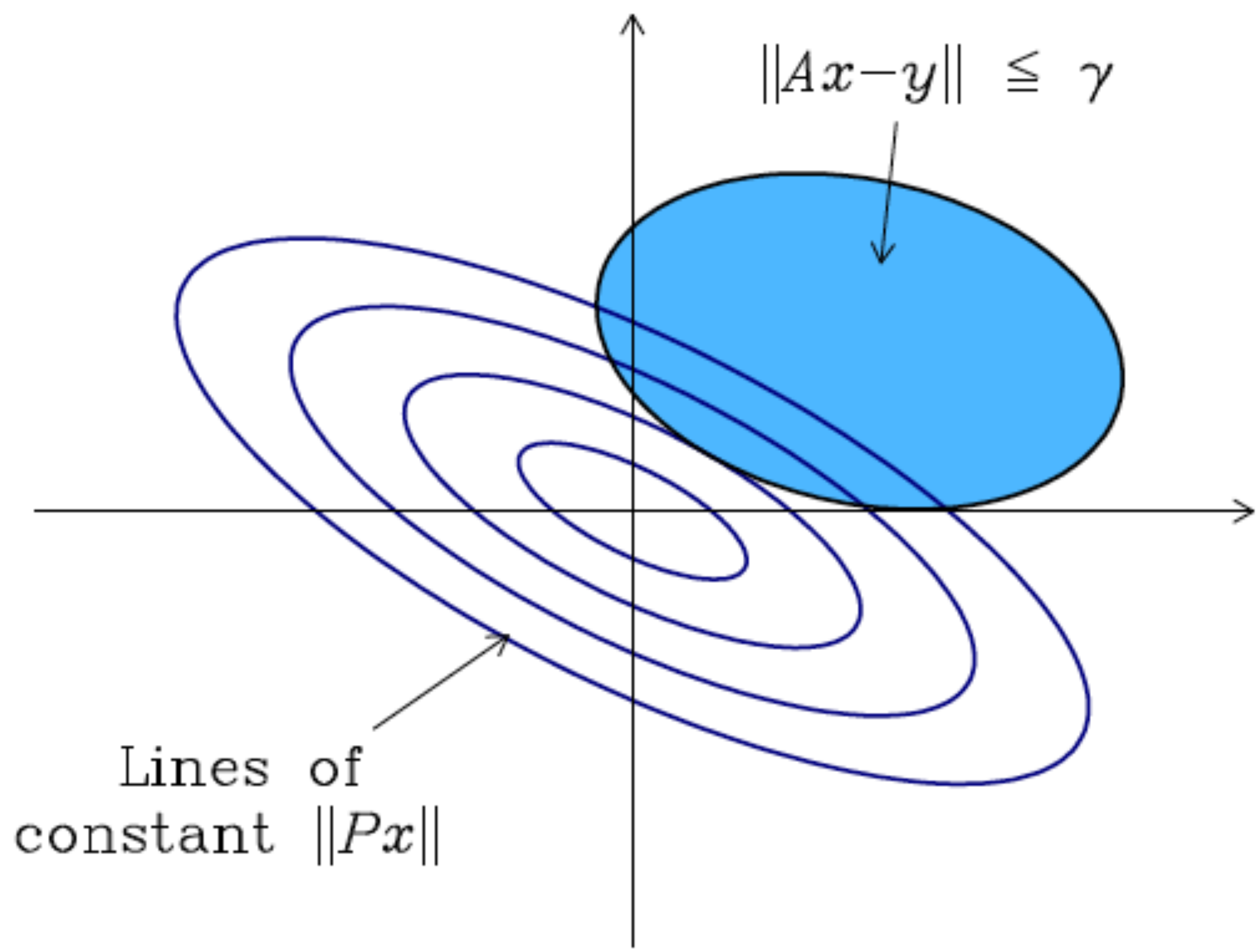

Figure 6.3: Minimization of $\|Px\|$ subject to $\|Ax-y\| \le \gamma$ for $x \in \mathbb{R}^2$.

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with $\mathbf{P} \in \mathbb{R}^{\mathbf{p}\times\mathbf{n}}$. $\mathbf{P}$ can be used to suppress undesirable properties in the model $\mathbf{x}$, e.g., slopes, magnitudes, curvature. Now our unconstrained function is

$$u(\mathbf{x}, \mu) = ||\mathbf{Px}||^{\mathbf{2}} - \mu(\gamma^{\mathbf{2}} - ||\mathbf{Ax} \; - \mathbf{y}||^{\mathbf{2}}) \tag{30}$$

Equivalently

$$u(\mathbf{x}, \mu) = ||\mathbf{Px}||^{\mathbf{2}} + \mu||\mathbf{Ax} \; - \mathbf{y}||^{\mathbf{2}} - \mu\gamma^{\mathbf{2}} \tag{31}$$

and we require $\mu > 0$ to minimize u. (31) represents a tradeoff between large $\mathbf{Px}$ and large data misfit. Decreasing misfit always increases the penalty norm $||\mathbf{Px}||$ and vice versa. Differentiating by μ recovers the equality constraint in (28), and derivatives on $\mathbf{x}$ don't see the γ term so we can just deal with

$$\hat{u}(\mathbf{x}) = ||\mathbf{Px}||^{\mathbf{2}} + \mu||\mathbf{Ax} \; - \mathbf{y}||^{\mathbf{2}} \tag{32}$$

UDLS with uncertainties - making the problem overdetermined

$$\hat{u}(\mathbf{x}) = ||\mathbf{Px}||^{\mathbf{2}} + \mu||\mathbf{Ax} \; - \mathbf{y}||^{\mathbf{2}} \tag{32}$$

$$= ||\mathbf{Px} - \mathbf{0}||^{\mathbf{2}} + ||\mu^{\frac{1}{2}}\mathbf{Ax} \; - \mu^{\frac{1}{2}}\mathbf{y}||^{\mathbf{2}} \tag{33}$$

Now we can write this as an overdetermined LS problem

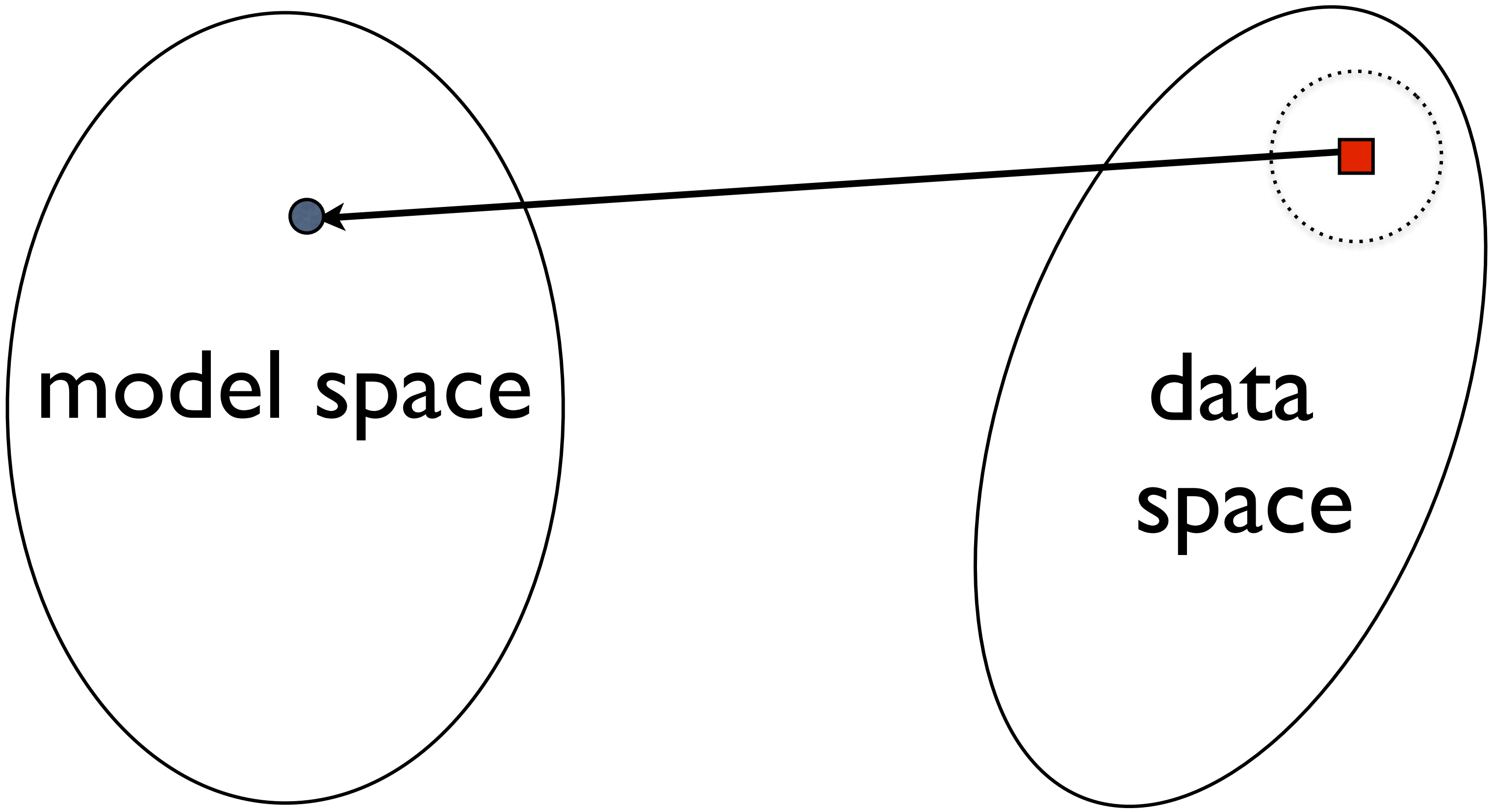
$$\hat{u}(\mathbf{x}) = \left|\left| \begin{bmatrix} \mathbf{P} \\ \mu^{\frac{1}{2}}\mathbf{A} \end{bmatrix} \mathbf{x} - \begin{bmatrix} 0 \\ \mu^{\frac{1}{2}}\mathbf{y} \end{bmatrix} \right|\right|^{\mathbf{2}} \tag{34}$$

$$= ||\mathbf{Cx} - \mathbf{d}||^{\mathbf{2}} \tag{35}$$

with $\mathbf{C} \in \mathbb{R}^{(\mathbf{p}+\mathbf{m})\times\mathbf{n}}$ and we need $p + m > n$.

We also need to find the value of μ to satisfy $||\mathbf{Ax} - \mathbf{y}|| \leq \gamma$ requiring multiple solutions of (35).

Inverse modeling:



Given real (observed) data $\mathbf{d} = (d_1, d_2, d_3, ..., d_M)$

with uncertainties $\sigma = (\sigma_1, \sigma_2, ..., \sigma_M)$

find an $\mathbf{m}$ to fit the data "well enough"

For noisy data (read: **all** data), we need a measure of how well a given model fits. Sum of squares is the venerable way:

$$\chi^2 = \sum_{i=1}^{M} \frac{1}{\sigma_i^2} \left[d_i - f(x_i, \mathbf{m}) \right]^2$$

or

$$\chi^2 = ||\mathbf{W}\mathbf{d} - \mathbf{W}\hat{\mathbf{d}}||^2$$

where $\mathbf{W}$ is a diagonal of reciprocal data errors

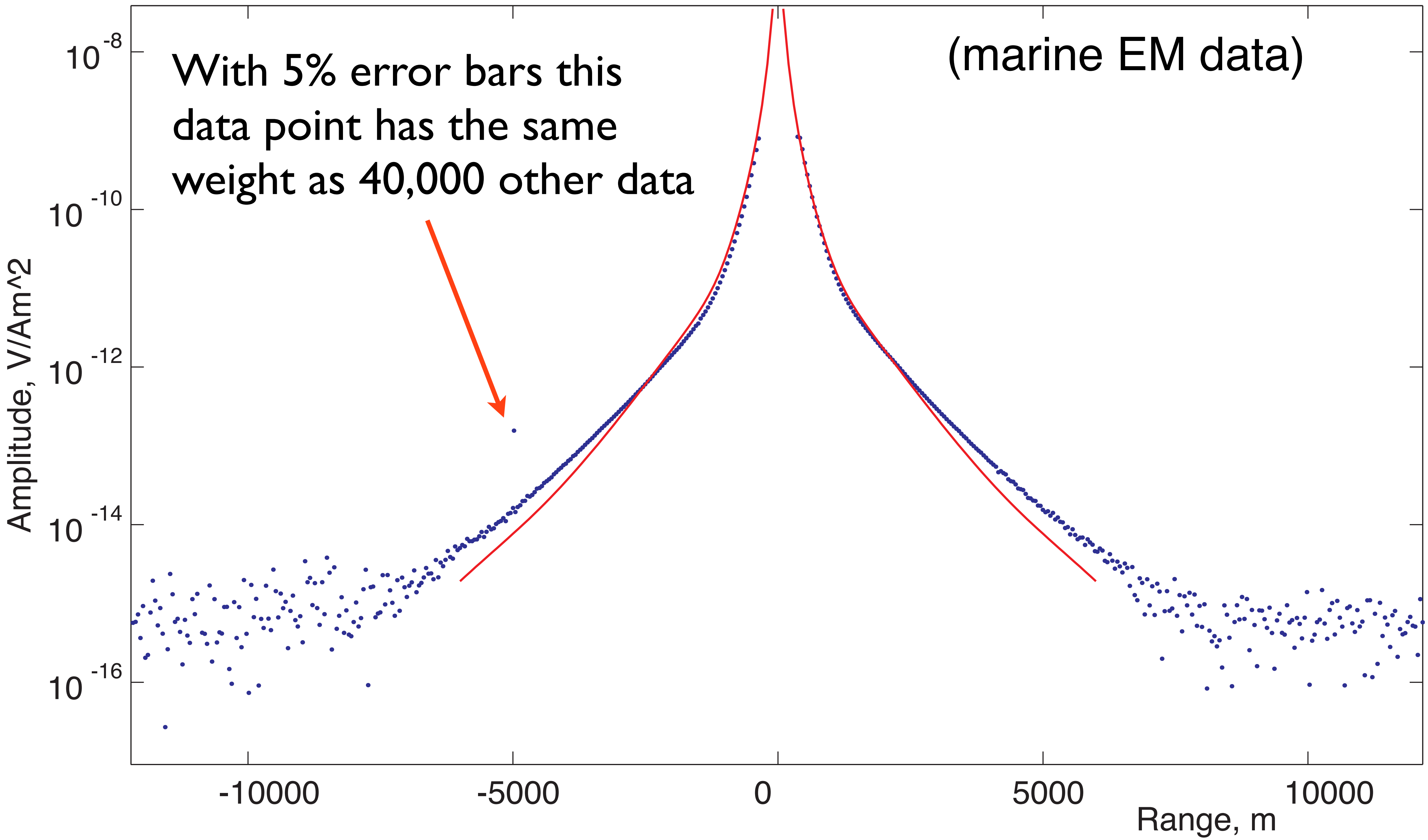
$$\mathbf{W} = \mathrm{diag}(1/\sigma_1, 1/\sigma_2, ..., 1/\sigma_M) \quad .$$

I like to remove the dependence on data number and use RMS:

$$\mathrm{RMS} = \sqrt{\chi^2/M} \quad .$$

We could use other measures of fit, but the quadratic measure works with the mathematics of minimization, and for Gaussian errors LS is a maximum likelihood, minimum variance, unbiased solution. But...

... sum-squared misfit measures are unforgiving of outliers:



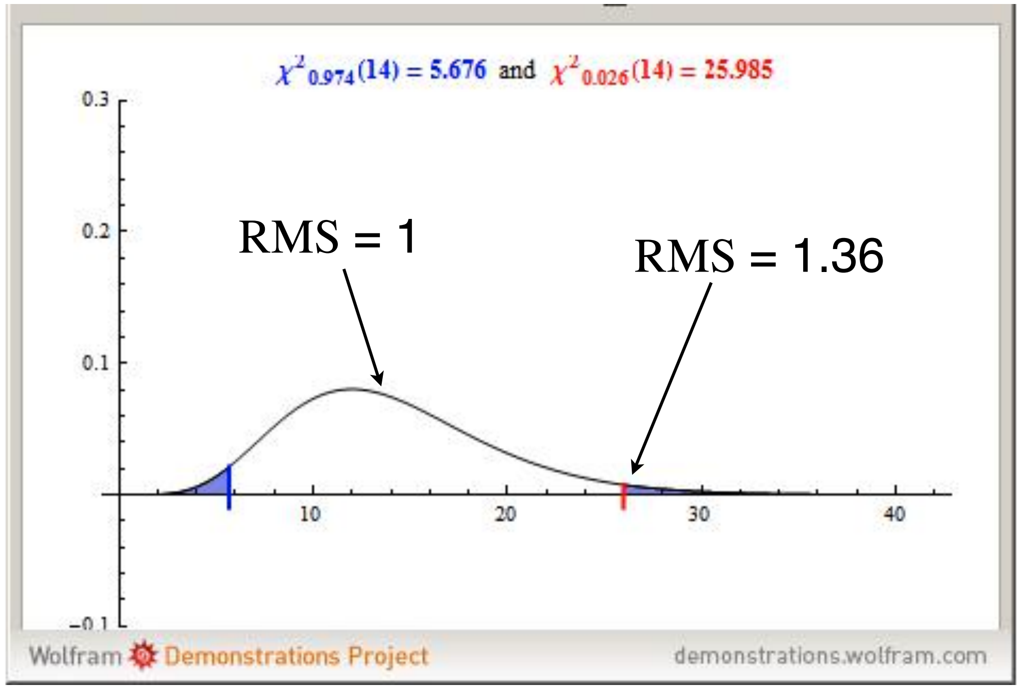
All through the inversion process you should monitor weighted residuals to ensure that there are no bad guys out there.

What value should we expect for χ^2_*?

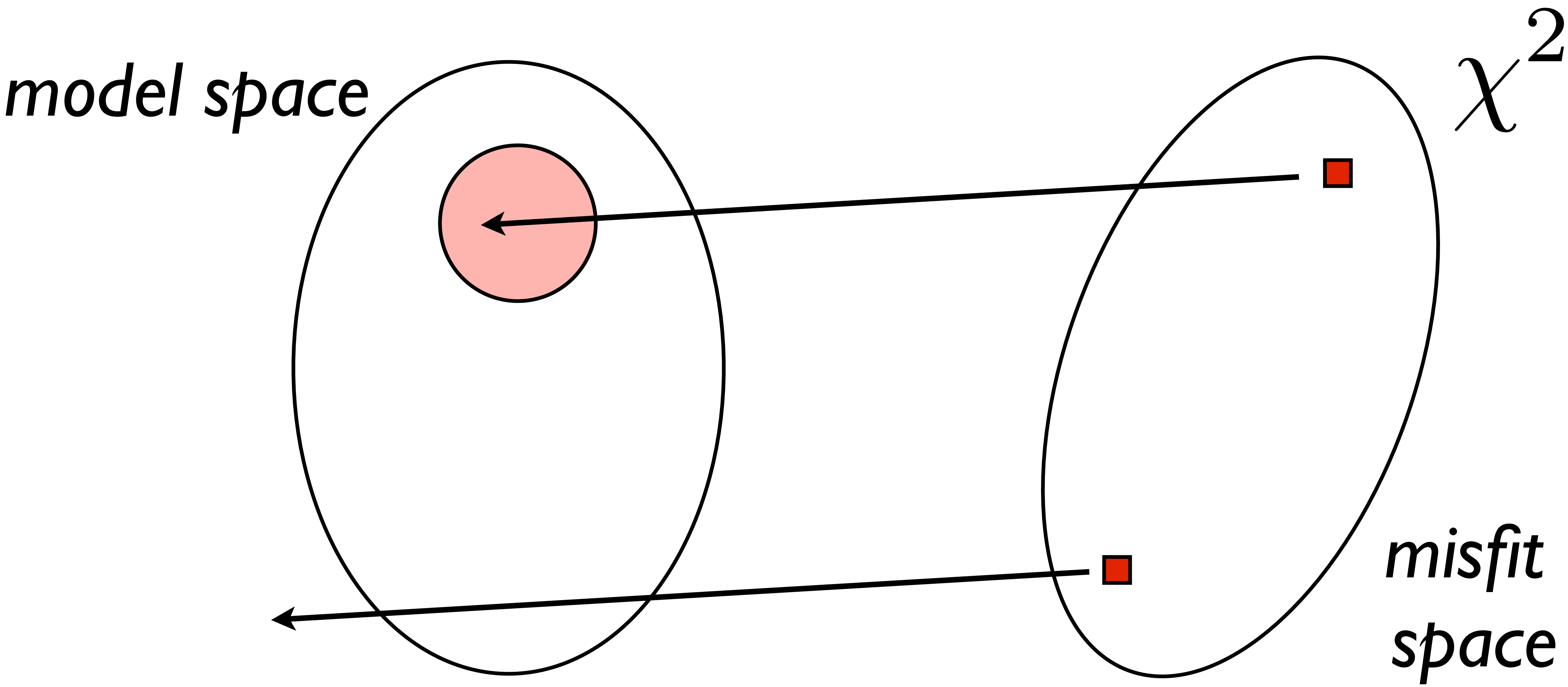
For zero-mean, Gaussian, independent errors, the sum-square misfit

$$\chi^2 = ||\mathbf{Wd} - \mathbf{W}\hat{\mathbf{d}}||^2$$

is chi-squared distributed with M degrees of freedom. The expectation value is just M, which corresponds to an RMS of one, and so this could be a reasonable target misfit. Or, one could look up the 95% (or other) confidence interval for chi-squared M.

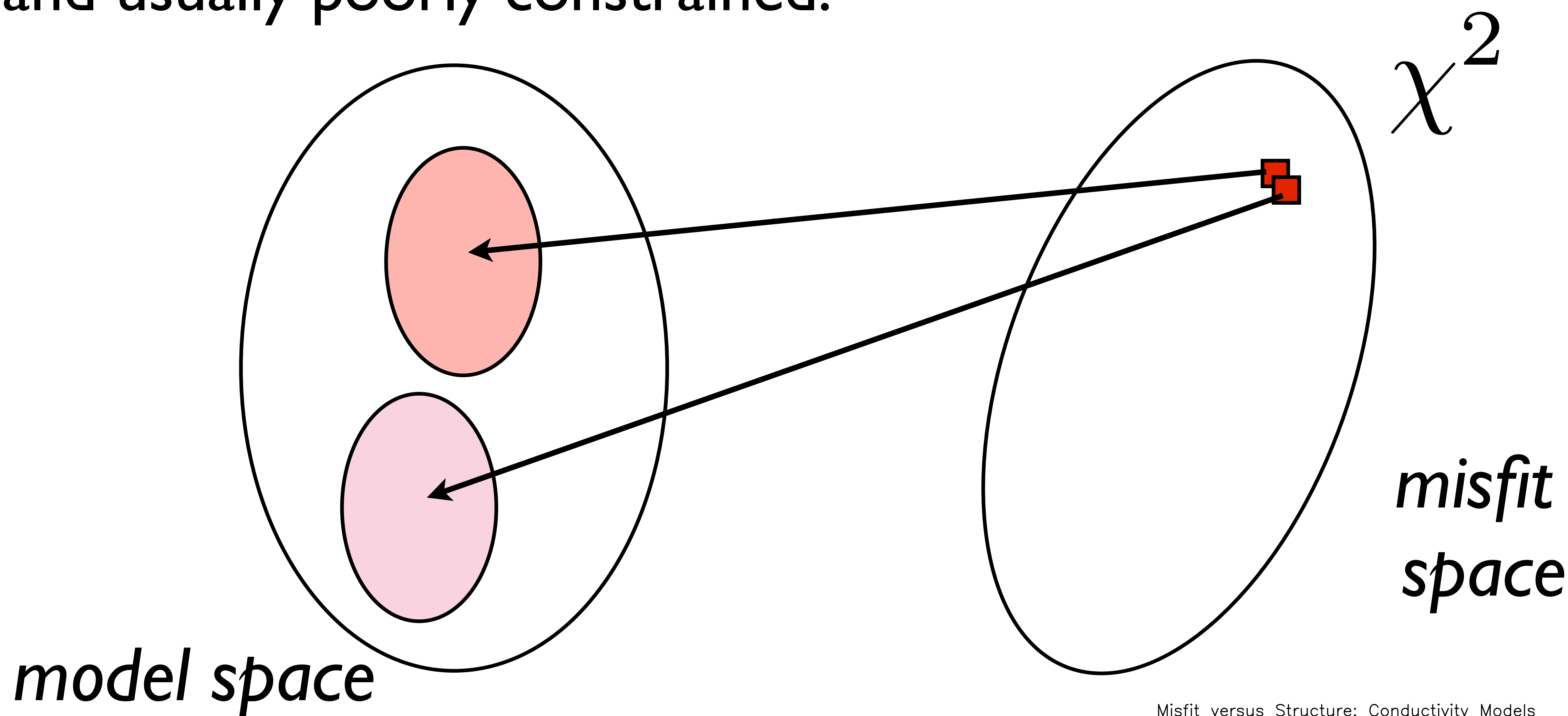


Geophysical inversion is non-unique:



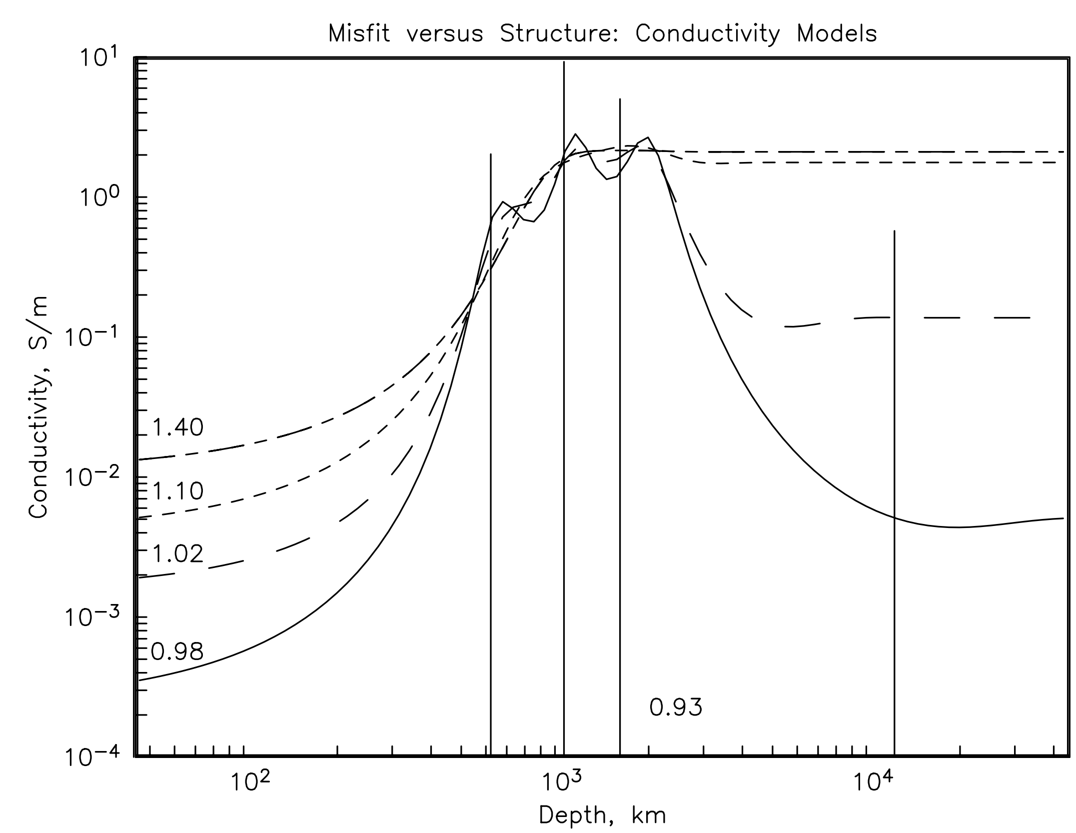
A single misfit will map into an infinite number of models (or none at all!).

and usually poorly constrained:



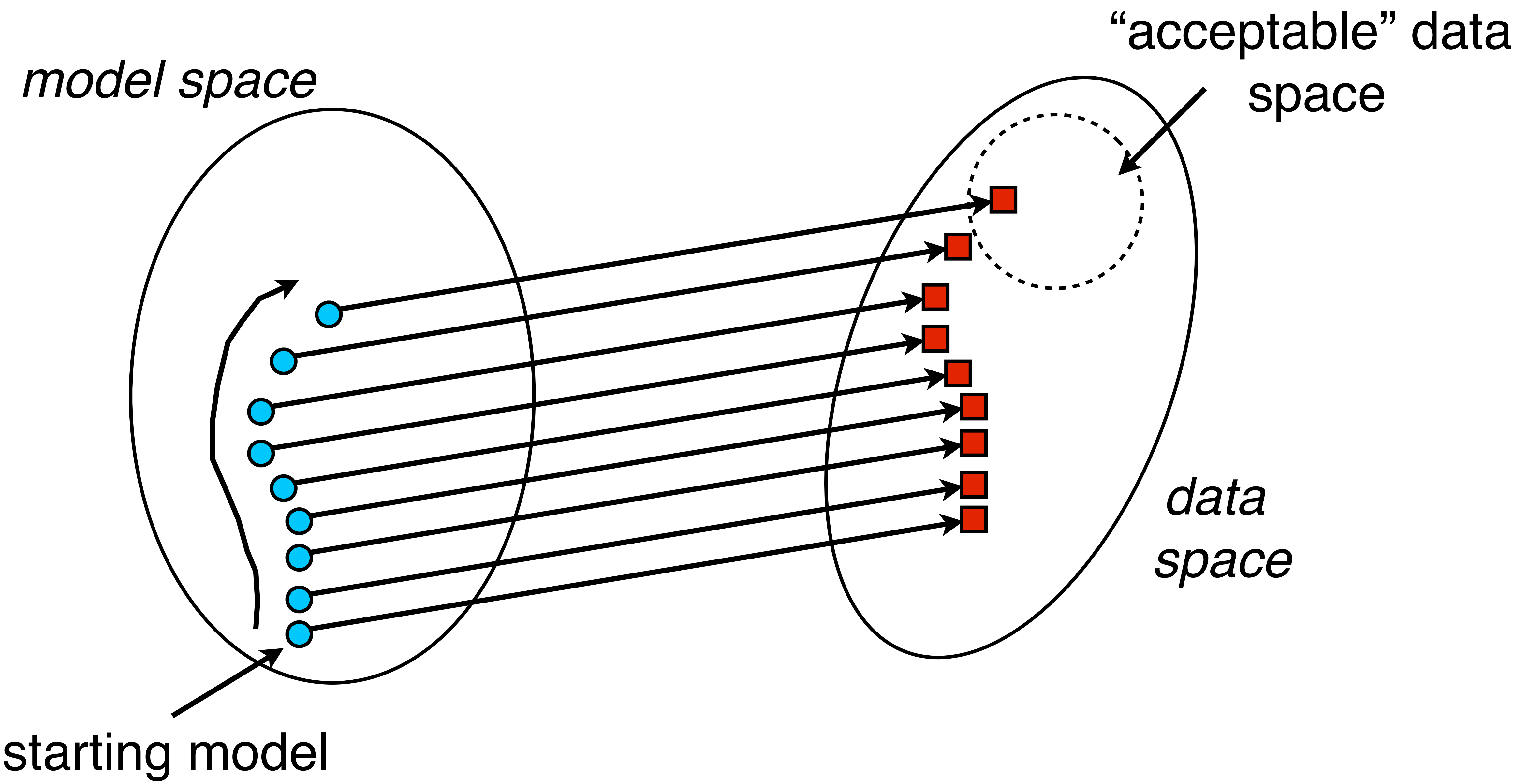
A small distance in χ^2 corresponds to a large distance in $\mathbf{m}$

(And don't forget: the minimum χ^2 is likely outside your model parameterization).

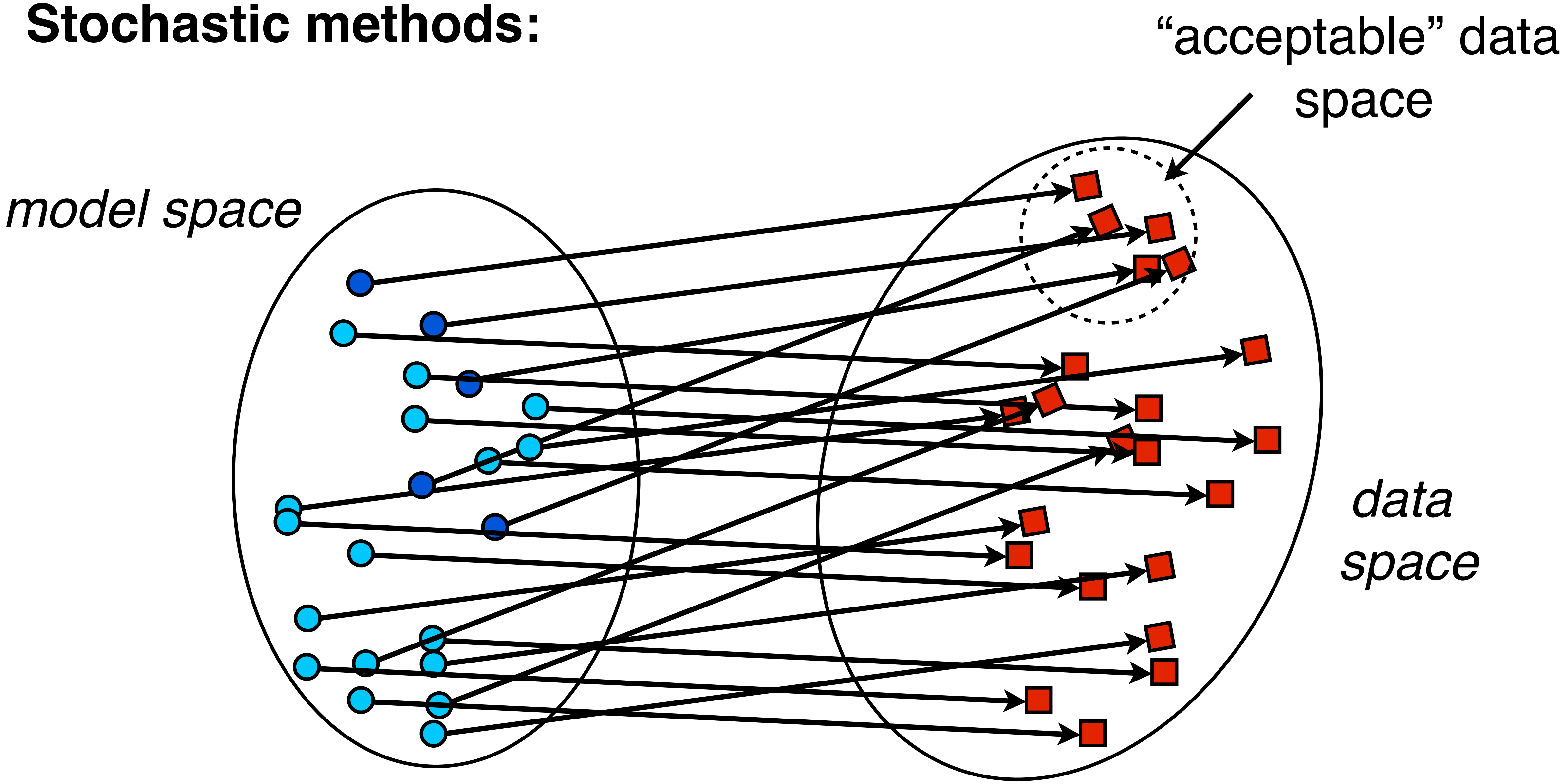


Deterministic

Newton Algorithms
Steepest descent
Conjugate Gradients



Stochastic methods:



A useful approach, largely restricted to simple problems (because millions of models required), with most of the subtlety in model generation methods.

The advantages are that (i) only forward calculations are made and (ii) some probabilities can be obtained on model parameters. Best for sparsely parameterized models. One needs to be careful that bounds on explored model space don't unduly influence the outcome.

Existence and Uniqueness: Is there a solution to the inverse problem? Is there only one solution?

Finite noisy data for a linear problem (say, gravity)
An infinite number of solutions fit the data

Finite noisy data for a nonlinear problem
Either zero or an infinite number of solutions fit the data

Infinite exact data
A unique solution has been shown to exist for a few cases.

As someone put it, there is no such thing as being a little bit non-unique